Time-Series Modeling of Narragansett Bay Fixed-Site Monitoring Stations' Network Data

A Machine Learning Approach

Siddharth Gupta
emailsiddha@gmail.com

ABSTRACT

Nutrient pollution poses a significant threat to Narragansett Bay, potentially causing hypoxic conditions that threaten the flora and fauna in the bay. To monitor the water body, the Rhode Island Department of Environmental Management has installed several monitoring stations at key bay locations. These stations monitor changes in metrics such as dissolved oxygen, salinity, and water temperature. In this project, several machine learning approaches have been used to model historical time-series data from these monitoring stations. Models were created to predict dissolved oxygen levels from other indicators and to forecast future dissolved oxygen levels for a 14-day period. Fitted models used tree-based architectures, neural networks (requiring imputation), and curve-fitting techniques. These models were compared using their MAPE (mean absolute percentage error) to labeled test data. Low MAPE in the best-performing models indicates that both tasks are viable. Using these models, an interactive web application was created, allowing the general public to visualize the complex relationships driving hypoxia in the bay: https://narrabaydata.streamlit.app/. Future work will explore more computationally intensive models and consider additional endogenous variables for longer-term modeling.

1 Introduction

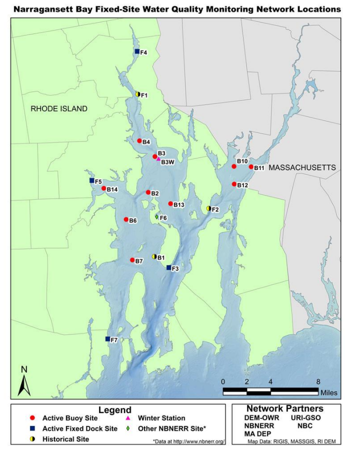
1.1 Narragansett Bay

Narragansett Bay is essential to Rhode Island, as it is home to thousands of species of plants and animals. The Bay contributes billions of dollars annually to Rhode Island's economy.[1] Part of what makes the Bay so productive is the fact that it is an estuary, where freshwater meets the sea, but this is also what makes it vulnerable to the effects of nutrient pollution. When the Bay is overly enriched with nutrients (a process known as eutrophication), this can cause issues such as algal blooms, in which excessive algae grow in the water and deplete the Bay's dissolved oxygen. This leads to a lack of dissolved oxygen, a condition known as hypoxia. Past literature on the Bay sets the hypoxia threshold at 5 mg/L, and this paper follows the same convention [1].

Hypoxia in the Bay has caused severe damage in the past. On August 20, 2003, a low dissolved oxygen (DO) event killed over one million fish in Greenwich Bay, a part of Narragansett Bay. Figure 1a shows the aftermath of the fish kill [2]. To control such events, the Rhode Island Department of Environmental Management has placed a series of buoy sensors across the bay that take 15-minute-interval measurements of key indicators, including DO, temperature, salinity, and more. Figure 1b shows a map of the sensors [3].

[1] https://dem.ri.gov/environmental-protection-bureau/water-resources/waters-wetlands/bay-and-coastal-waters/introduction

(a) Aftermath of the 2003 Greenwich Bay fish kill [2].



(b) Map of fixed-site monitoring stations [3].

Figure 1: Motivation and monitoring network.

1.2 Hypoxia Modeling

Past efforts to model hypoxia have focused on developing mechanistic models relying on physical phenomena. While such models are often accurate and precise, they can also be challenging to calibrate [4]. As such, a new paradigm of using statistical models is effective while requiring less calibration, provided that high-quality data is provided for modeling [4, 5]. Effective models include tree-based and neural network models. This project aims to develop such models of hypoxia in Narragansett Bay.

2 Project Description

2.1 Data Processing

Data was obtained in a single, clean Excel file. The Pandas library in Python was used for preprocessing.

2.2 Imputation

As multi-layer perceptrons, the neural network architecture used in this project, cannot train on missing data, missing data was imputed to train only that prediction model. Data "missingness" is shown in Figure 2. The MICE imputation framework was used to impute data.

Figure 2: Data availability by specific variable. Bars show the fraction of non-missing observations (left axis); labels above each bar give the raw count of available observations (right axis).

2.3 Modeling

Two different kinds of models were fit: those to predict DO levels from other indicators ("prediction"), and those to forecast future dissolved oxygen levels ("forecasting"). Models leveraged include the tree-based Random Forest, XGBoost, ExtraTrees, and HistGradientBoosting for prediction; the neural network multi-layer perceptron for prediction; and the time-series forecasting models Facebook Prophet, XGBoost (implemented for time-series), Exponential Smoothing, and AutoARIMA [6–9]. Tree-based models were chosen because they are resilient to missing data (as decision trees learn split directions during training), and a neural network was selected to capture potentially nonlinear, complex relationships in the data. The scikit-learn, XGBoost, and Darts libraries were used.

3 Results

Models were compared based on their Mean Absolute Percentage Error (MAPE), defined over n test observations with true values y_i and predictions $\hat{y}_i$ as

$$\mathrm{MAPE} = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| . \tag{1}$$

MAPE results are shown in Table 1 and Table 2.

Model	Imputation?	MAPE (bottom DO)	MAPE (surface DO)
MLPRegressor	**Required**	**4.67%**	**3.49%**
HistGradientBoosting	Not required	9.61%	4.76%
ExtraTreeRegressor	Not required	10.03%	4.85%
XGBoost	Not required	9.89%	5.00%
Random Forest	Not required	10.03%	5.04%

Table 1: DO prediction model comparison. Best-performing model in bold.

Model	Covariates?	Imputation?	GSO Dock MAPE (14-day forecast)
Facebook Prophet	**Y**	**Not required**	**5.02%**
XGBoost	Y	Required	7.52%
Exponential Smoothing	N	Required	7.86%
AutoARIMA	N	Required	8.12%

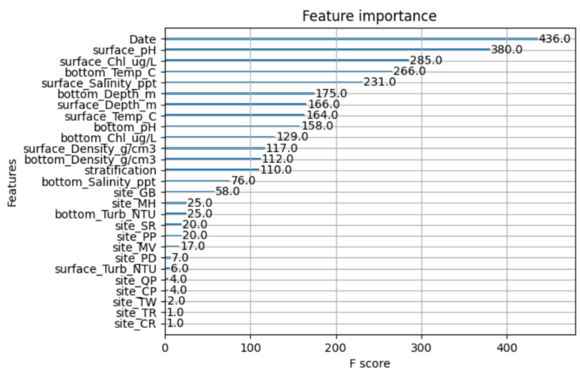
Table 2: 14-day DO forecasting model comparison at the GSO Dock station. Best-performing model in bold.

For the prediction task, the multi-layer perceptron was the strongest model, achieving 3.49% MAPE on surface DO and 4.67% on bottom DO — roughly halving the bottom-DO error of the best tree-based model (HistGradientBoosting, 9.61%). This advantage came at the cost of requiring imputed data, whereas the tree-based models were trained directly on data with missing values. Notably, all models found bottom DO harder to predict than surface DO, with a potential explanation being the complex dynamics (stratification, benthic processes) governing oxygen at depth. For the forecasting task, Facebook Prophet delivered the lowest error (5.02% MAPE over a 14-day horizon), outperforming the XGBoost forecaster (7.52%) and both univariate baselines. Prophet's advantage seems to be consistent with its additive decomposition of the signal,

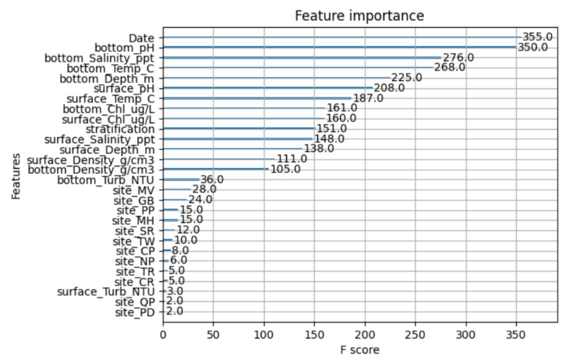
$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t , \tag{2}$$

into trend $g(t)$, periodic seasonality $s(t)$, holiday effects $h(t)$, and noise ε_t [9]: dissolved oxygen at GSO Dock is dominated by annual seasonality and is noisy.

Feature importance was calculated for the XGBoost model for prediction for both surface and bottom DO (Figure 3). The model strongly weights variables such as date (which makes sense, given strong seasonal patterns), pH, salinity, and temperature.



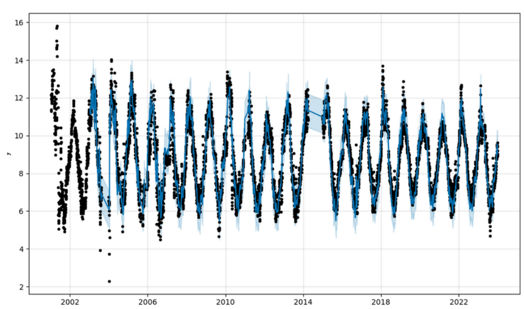
(a) Surface DO.



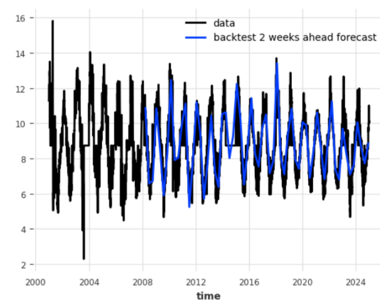
(b) Bottom DO.

Figure 3: XGBoost feature importance for DO prediction.

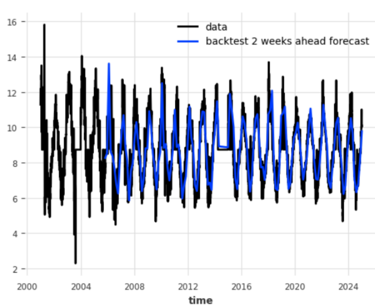
Two-week backtest graphs of the forecasting models were also created (Figure 4). An interactive visualization website with these models and interactive graphs was also created (https://narrabaydata.streamlit.app/).



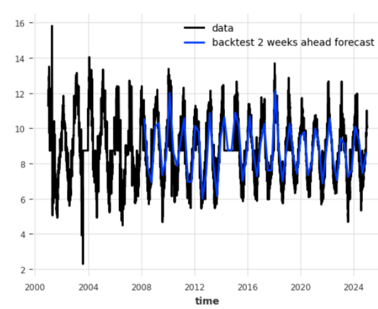
(a) Prophet.



(b) Exponential Smoothing.



(c) XGBoost.



(d) AutoARIMA.

Figure 4: Two-week backtests of the forecasting models at the GSO Dock station. Black: observed dissolved oxygen; blue: two-weeks-ahead backtest forecast.

4 Discussion

MLP outperforming tree-based models

The gap was largest for bottom DO (4.67% vs. 9.61%+). One plausible explanation is that bottom DO depends on smooth, and interacting physical phenomena, such as temperature, salinity-driven stratification, and seasonal cycles. A neural network can represent these continuously. while tree ensembles must approximate them. It is also possible that part of the MLP's advantage came from imputation itself: MICE-completed inputs may have given the network better priors.

Feature importance

The XGBoost importances (Figure 3) rank date, pH, salinity, and temperature highly. Each has a clear environmental interpretation: date encodes the strong seasonal cycle of hypoxia (peaking in stratified summer months); pH covaries with DO because photosynthesis and respiration are obviously correlated; and temperature and salinity both control density stratification, which limits the supply of oxygen to bottom waters. Seawater density is a function of temperature, salinity, and pressure. Thus, when warm, fresh water overlies cold, salty water, the resulting density gradient suppresses vertical mixing and starves bottom waters of oxygen. In these conditions, the temperature and salinity features are interpretable.

Limitations.

Forecasting results are reported for a single continuously-monitored station (GSO Dock) and thus may not generalize to stations with sparser records. Additionally, the models are correlational: they cannot attribute hypoxia to nutrient loading or predict the effects of potential interventions, and are best used alongside mechanistic models rather than in place of them.

5 Conclusion

Low MAPE scores indicate that statistical modeling for Narragansett Bay is viable, and that forecasting is viable for sites that take continuous data. The success of statistical models in predicting Narragansett Bay hypoxia (and hypoxia elsewhere) shows that they hold promise for the future of environmental stewardship. Such models may be used in tandem with existing mechanistic models to improve the quality and speed of responses to hypoxia. Looking ahead, there is significant potential for further work on this topic. This project was conducted without access to large amounts of compute, so it would be possible to train larger, better models with more GPU time. Additionally, to improve the models, considering additional covariates, such as river flow, weather forecasts, or other historical data related to the Bay, could be helpful. The models created could be implemented conservatively with other models in a real-world context to evaluate performance "in the wild."

Acknowledgment

All monitoring data used in this project was made available upon request from the Rhode Island Department of Environmental Management's Fixed-Site Monitoring Stations Network. The author thanks RIDEM and the Narragansett Bay Fixed-Site Monitoring Network partners for maintaining this dataset, without which this work would not have been possible.

References

[1] Dorothy Q. Kellogg. Hypoxia in Narragansett Bay: Fixed-site data analyses. Technical report, Rhode Island Department of Environmental Management, April 2018. URL https://www.rimonitoring.org/ws/wp-content/uploads/2018/05/Hypoxia_NarrBay_Fixed_Site_Data_Analyses_April2018.pdf.

[2] Rhode Island Environmental Monitoring Collaborative. Dissolved oxygen – recent trends. https://www.rimonitoring.org/dissolved-oxygen/recent-trends/. Accessed 16 Mar. 2025.

[3] Rhode Island Department of Environmental Management. Fixed-site monitoring stations network data. https://dem.ri.gov/environmental-protection-bureau/water-resources/research-monitoring/narraganset-bay-assessment-data. Accessed 6 Feb. 2025.

[4] Dimitris V. Politikos, Georgios Petasis, and George Katselis. Interpretable machine learning to forecast hypoxia in a lagoon. *Ecological Informatics*, 66:101480, December 2021. doi: 10.1016/j.ecoinf.2021.101480.

[5] Xin Yu, Jian Shen, and Jiabi Du. A machine-learning-based model for water quality in coastal waters, taking dissolved oxygen and hypoxia in Chesapeake Bay as an example. *Water Resources Research*, 56 (9):e2020WR027227, 2020. doi: 10.1029/2020WR027227.

[6] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.

[7] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016. doi: 10.48550/arXiv.1603.02754.

[8] Hind Taud and Jean-François Mas. Multilayer perceptron (MLP). In *Geomatic Approaches for Modeling Land Change Scenarios*, pages 451–455. Springer, 2018.

[9] Sean J. Taylor and Benjamin Letham. Forecasting at scale. *The American Statistician*, 72(1):37–45, 2018. doi: 10.1080/00031305.2017.1380080.